

What a self-playing population learns, what it forgets, and which of the two you measure

Roland Loechli

Draft, October 3, 2026

Abstract

In a self-play genetic algorithm, the individual exported as “the champion” is close to a median member of its own population, and much of what looks like coevolutionary instability in its learning curve is noise added by the rule that picks it. We show this for David Ha’s tournament-selection algorithm on Slime Volleyball, run as a designed experiment: 59 independent runs of 500,000 self-play games in a compiled environment that reproduces the reference implementation bit for bit, every agent scored against a frozen opponent never seen in training. Ha’s rule promotes the longest winning lineage; that individual ranks on average 60 of 128 in its population, its lineage counter is uncorrelated with skill ($\rho = +0.06$), and the population is not cycling. Exporting instead the winner of a short internal tournament did better in a preregistered test on fresh runs, though by less than a post hoc analysis had suggested. Internal improvement precedes external transfer by tens of thousands of games, and the timing of the phase change varies 7.5-fold across seeds. An archive of past champions used as a parent abolishes learning; used as a test, it neither harms nor helps detectably, nor does one organised by behaviour. These claims held in a preregistered replication on 36 fresh runs, and the export-rule failure persists in a game with tunable cyclic skill, where the better rule helps less. Algorithm families reach a similar ceiling, and an apparent difference in their reliability did not survive fresh seeds; NEAT, in one configuration, never learned. A two-population variant with unequal capacity is exploratory. Any evolutionary loop that promotes one artefact out of a population, including LLM-driven program evolution, has the same exposure; we state this as a hypothesis and give the diagnostic.

1 Introduction

Self-play evolution produces competent agents from an entirely internal signal. In the steady-state genetic algorithm that Ha (2020) uses for Slime Volleyball, two members of a population of 128 play one game, the loser is overwritten by a mutated copy of the winner, and that is all: nothing in the loop refers to an absolute standard of play. Run for 500,000 games, it produces an agent that holds the expert policy shipped with the environment to a draw. The learning curve by which this is usually reported, the score of an exported champion against that expert, is also strikingly unstable: competence is reached, lost and regained between adjacent checkpoints.

The textbook explanation is coevolutionary forgetting. Fitness is relative, so a skill that no current opponent punishes is not selected for and can be lost, and the population cycles (Cliff and Miller, 1995; Popovici et al., 2012); the standard remedy is a hall of fame of past champions to test against (Rosin and Belew, 1997). This paper asks where the instability actually lives, and answers with a design rather than a run: 59 runs across conditions that each change one thing, a pre-registered evaluation protocol, and measurements the external learning curve cannot make.

The main finding is that the explanation is wrong here and the instability is mostly a reporting artefact. Ha’s algorithm exports the individual with the longest winning lineage, by design “without

actually computing who is best to save time” (Ha, 2020). Because the lineage counter is inherited by the loser on every replacement, it measures the age of a lineage, not the merit of an individual. Scored against the whole population, the exported champion is a middling member of it, and the population contains individuals 0.63 points per episode better on held-out episodes. Meanwhile a round robin of every checkpoint against every other shows skill to be almost perfectly transitive: later champions beat earlier ones, and the population is not going in circles. In Miconi’s (2009) terms, historical progress is steady; what fluctuates is our estimate of global progress, and it fluctuates because of which individual we chose to measure.

Contributions.

1. The export rule, not coevolution, is the dominant noise source in the reported curve; a cheap internal tournament exports measurably better champions in a preregistered test on fresh runs, recovering 43% of the gap to the population’s best member (§3.4).
2. Internal improvement precedes external transfer by tens of thousands of games, and the phase change is robust in occurrence but not in timing (§3.1, §3.2).
3. An archive used as a parent abolishes learning; used as a test it is indistinguishable from no archive, and the archive’s own win rate shows why (§3.6).
4. Four algorithm families reach a similar ceiling; the difference in how reliably they reach it that six seeds suggested did not survive fresh control runs (§3.7).
5. The findings hold in a preregistered replication on fresh seeds (§3.8); the export-rule failure persists when skill is made cyclic, and neither a behaviour-organised archive nor a topology-evolving algorithm changes the picture (§3.9).
6. An exploratory two-population block with unequal capacity (§4).
7. A compiled port of the environment, validated bit for bit, and an analysis pipeline in which every number is generated and checked against the raw files (§7).

Related work. Measuring progress in coevolution is an old problem: Cliff and Miller (1995) introduced tests of current individuals against ancestral opponents, Miconi (2009) separated local, historical and global progress, and recent work compares archive-based methods for global progress (Simione and Nolfi, 2020; Nolfi and Pagliuca, 2025); in fully enumerated number games, Seals and Tauritz (2026) find that the fitness approximations of competitive coevolution rank solutions deceptively enough to break elitism, more so with a hall of fame, with more intransitivity and with stronger survival selection. The export rule studied here is such an approximation at the point of reporting, in a game too large to enumerate. Memory mechanisms range from the hall of fame (Rosin and Belew, 1997) to Pareto and Nash-based archives (Ficici and Pollack, 2001, 2003); disengagement between unequal populations is studied by Cartlidge and Bullock (2004). In symmetric zero-sum games, any game decomposes into a transitive and a cyclic part (Balduzzi et al., 2019), and real games appear to be transitive at the extremes of skill with cycles concentrated in between (Czarnecki et al., 2020). On the evaluation side, single-seed results in reinforcement learning have repeatedly been shown to mislead (Henderson et al., 2018; Agarwal et al., 2021). What we add is a measurement of the step that all of these take as given: which individual is reported.

2 Setup

Task. Slime Volleyball is a two-player, zero-sum, symmetric game (Ha, 2015), used here through `slimevolleygym` (Ha, 2020). Each agent observes a 12-dimensional state in its own frame and presses 3 binary buttons. An episode ends when a side loses 5 lives or after 3,000 steps; its score is points won minus points lost, in $[-5, 5]$. Every evolved agent is the same fixed 12–10–10–3 tanh network with 273 parameters. Nothing but the weights evolves.

External yardstick. The environment ships a small recurrent policy evolved by Ha (2015): 112 parameters, whose 15 inputs are 8 game observations (its own and the ball’s state, never its opponent’s) and its own previous outputs. It is never used in training. Every checkpoint is scored against it over 200 episodes on one evaluation seed; the final and the best checkpoint of each run are re-scored over 1,000 episodes on a disjoint seed, because the best of 100 checkpoints is a selected maximum. All headline numbers are held out. The protocol was fixed before the runs and is recorded with the data.

The control algorithm. A faithful replication of Ha’s tournament-selection GA: population 128, initial weights $\mathcal{N}(0, 0.5^2)$, mutation $\sigma = 0.1$. Each of 500,000 games draws two members; the loser is replaced by a mutant of the winner and inherits the winner’s streak counter, which then grows by one; a tie mutates one player in place. Every 5,000 games the member with the largest streak is exported, giving 100 checkpoints per run.

Conditions. Each condition changes one thing (Table 1). Seed counts differ by design: 6 for every condition that carries a claim, fewer for side conditions that were cut when the design was revised. Every change after the first run started is recorded, with what was known at the time, in a decision log that ships with the data.

Measurements. Beyond the learning curve, three measurements need no external opponent: (i) a round robin of a run’s checkpoints, spaced 50,000 games apart, with Bradley–Terry ratings (Bradley and Terry, 1952) on the Elo scale and a count of cyclic triads; (ii) for the control, which snapshots its whole population every 50,000 games, every member is scored against the yardstick and compared with the exported one; (iii) the internal transition, the first point at which the population’s own training games last more than 1,500 steps. Comparisons use the exact two-sided Mann–Whitney test with full enumeration (Mann and Whitney, 1947) and Cliff’s δ as effect size; with 6 runs a side the smallest attainable p is 0.002. Secondary comparisons are uncorrected for multiplicity. Because a run that never learns has no volatility, every stability comparison is also reported over the runs that learned.

3 Results

The results follow one line. What the population learns, and when (§3.1–3.3); what the reported curve measures instead, and a better rule for choosing the champion (§3.4); what does not explain the instability, and what archives and other algorithm families change (§3.5–3.7); and whether all of it holds, on fresh seeds and beyond this game, archive and topology (§3.8, §3.9).

Table 1: Conditions analysed. Two-population runs count once per seed.

condition	what it changes relative to the control	runs
control	Ha’s 2020 GA: population 128, $\sigma = 0.1$, champion = longest winning streak	6
hof-eval-v2	archive as test: with $p = 0.25$ the opponent is an archived champion (capacity 512); genes come only from the living pool	6
hof-0.25	archive as parent: with $p = 0.25$ the opponent is an archived champion (capacity 64) whose mutant replaces a member it beats	6
hof-0.50	archive as parent: with $p = 0.5$ the opponent is an archived champion (capacity 64) whose mutant replaces a member it beats	1
hof-full	archive as parent: with $p = 0.25$ the opponent is an archived champion (capacity 512) whose mutant replaces a member it beats	3
ga2015	generational GA: population 100, 10 games per individual per generation, top 20 kept, uniform crossover, $\sigma = 0.1$	6
es	self-play OpenAI-ES: 50 mirrored perturbations, 4 games each, learning rate 0.03, $\sigma = 0.1$; reports the mean	6
sigma-0.05	mutation scale $\sigma = 0.05$	3
sigma-0.20	mutation scale $\sigma = 0.2$	3
pop-32	population 32	1
asym1x	two populations of 128, 273 v 273 parameters, $\sigma = 0.100$ v 0.100 ; a share 0.25 of games crosses populations	6
asym2x	two populations of 128, 531 v 273 parameters, $\sigma = 0.100$ v 0.100 ; a share 0.25 of games crosses populations	6
asym2x-norm	two populations of 128, 531 v 273 parameters, $\sigma = 0.072$ v 0.100 ; a share 0.25 of games crosses populations	6

Table 2: Headline results. Points per episode against the 2015 policy, mean $\pm$ s.e.m. across runs, held out. “Learned to rally”: the population ever held 1,500-step rallies against itself. The last two rows are the single run on the unmodified environment and Ha’s published value.

condition	runs	learned to rally	best champion (held out)	end-of-run champion	checkpoints above parity
control (Ha 2020 GA)	6	6/6	$+0.32 \pm 0.06$	-0.15 ± 0.27	26%
archive as test, full span	6	5/6	-0.55 ± 0.86	-1.01 ± 0.83	18%
archive as parent, $p=0.25$	6	1/6	-3.94 ± 0.90	-4.10 ± 0.74	2%
archive as parent, $p=0.50$	1	0/1	$-4.84 \pm \text{—}$	$-4.84 \pm \text{—}$	0%
archive as parent, full span	3	0/3	-4.83 ± 0.01	-4.84 ± 0.00	0%
generational GA (Ha 2015)	6	5/6	-0.68 ± 0.83	-2.00 ± 0.82	3%
self-play ES	6	4/6	-1.56 ± 0.99	-2.08 ± 0.94	7%
sigma = 0.05	3	3/3	$+0.34 \pm 0.07$	-0.17 ± 0.23	26%
sigma = 0.20	3	2/3	-1.44 ± 1.70	-1.66 ± 1.60	11%
population 32	1	0/1	$-4.84 \pm \text{—}$	$-4.86 \pm \text{—}$	0%
<i>reference run, unmodified environment</i>	1	1/1	$+0.50 \pm 0.03$	$+0.04 \pm 0.02$	22%
<i>Ha (2020), same algorithm and budget</i>	1	—	$+0.35 \pm 0.02$	—	—

3.1 Internal improvement precedes external transfer

The reference run, Ha’s algorithm at Ha’s budget on the unmodified environment, has two regimes (Figure 1). At first its champion loses almost every point to the 2015 policy; then it climbs steeply and starts producing champions that beat it. The internal signal moves first: the population’s own rallies pass 1,500 steps at 104,200 games, and the first champion above parity appears at 172,000, a lag of 67,800 games. An evaluation stopped in between reports total failure from a population that is already improving. Competence here means not losing: after the transition the modal outcome is a draw at the step limit, and the final champion wins 19%, draws 64% and loses 17% of held-out episodes (Appendix A.1). Held out, the best checkpoint scores +0.50 and the final one +0.04, against Ha’s published +0.35 for the same algorithm and budget.

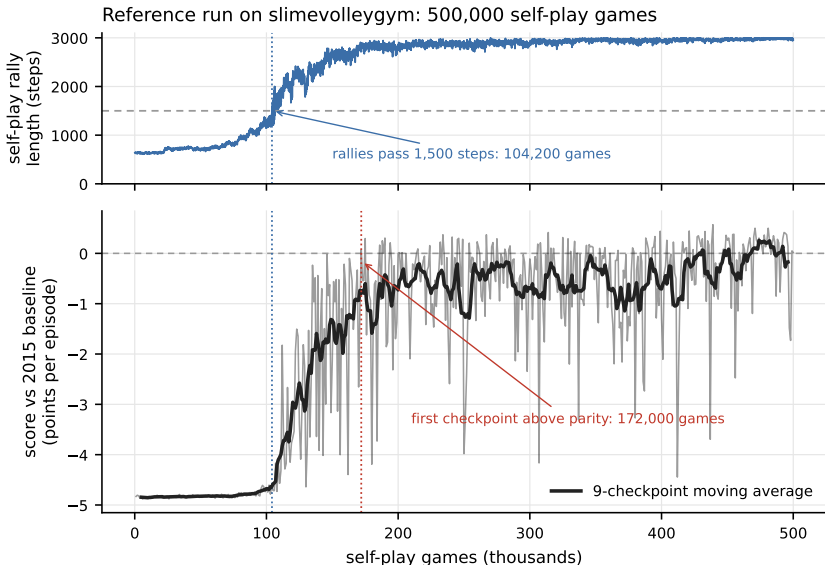


Figure 1: The reference run. Top: rally length of the population’s training games against itself. Bottom: score of the exported champion against the 2015 policy, every checkpoint (thin) and a moving average (thick). Dotted lines mark the internal transition and the first checkpoint above parity.

3.2 The phase change is robust; its timing is not

Across 6 control seeds the internal transition happens anywhere from 55,000 to 415,000 games, a 7.5-fold spread, and the first above-parity checkpoint from 85,000 to 440,000 (Figure 2). The lag between them is always present, 25,000 to 100,000 games. Every seed reaches the same band (6/6 learned to rally, 6/6 produced an above-parity champion), so the occurrence of the transition replicates and its timing does not. A single run supports no claim about when self-play starts working, although the first version of this study, with one seed, reported a single transition time. Nor does the swing settle once the level rises: in 100,000-game windows the control’s mean climbs from -4.32 to -0.32 , while the spread of checkpoint scores within a run stays between 0.51 and 0.83 (Appendix A.1).

3.3 The population is not cycling

If the swings were forgetting, later champions should not reliably beat earlier ones, and the checkpoint tournament should contain cycles. Neither is the case (Figure 3). Within control runs the rank

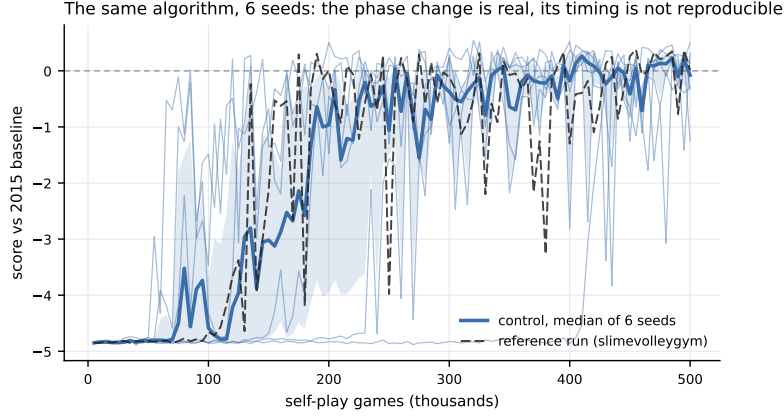


Figure 2: Every control seed (thin), median and inter-quartile band (thick), and the reference run on the same checkpoint grid (dashed).

correlation between Elo and training time averages $+0.74$, and 0.2% of decided checkpoint triads are cyclic; no condition whose runs all learned exceeds 0.2% . Only the archive-as-parent runs, which never learned, show substantial intransitivity: their champions are equally bad policies whose pairwise results are noise. The game, at the scale of skill explored here, is dominated by its transitive component (Balduzzi et al., 2019; Czarnecki et al., 2020).

The checkpoints are 50,000 games apart, so a faster cycle would be invisible. Played at 5,000-game spacing (all 100 champions of each control run, 20 games per pair; Appendix A.7), the same deadband rule finds 1.0% of decided triads cyclic, but at that sample size the deadband lets noise decide pairs between near-equal policies. When a pair counts as decided only if an exact sign test on its wins and losses says so, 35 of 273,376 decided triads are cyclic (0.01%): not cycling at the finer resolution either.

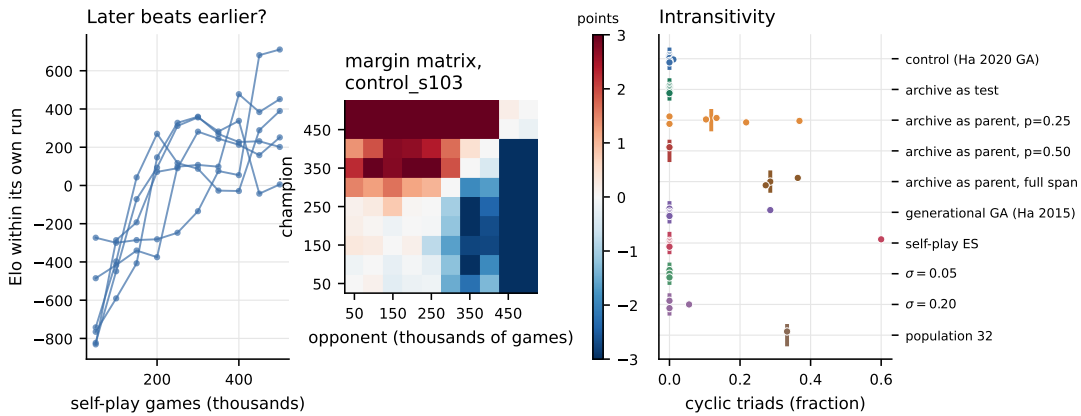


Figure 3: Left: Elo of each checkpoint within its own run, control seeds. Middle: pairwise margin matrix of one control run, later checkpoints against earlier. Right: share of decided checkpoint triads that are cyclic, per condition.

3.4 The export rule is the noise source

Because the control snapshots its whole population, the exported champion can be compared with every individual it was chosen from (Figure 5), each scored on 30 episodes against the 2015 policy; once skill varies within the pool, those scores agree with an independent measurement at a median $r = 0.94$. Over 6 seeds and all snapshots, the exported individual ranks on average 60 of 128 in its own pool, and the rank correlation between the streak counter and actual skill is $+0.06$. Re-scored on 1,000 held-out episodes, the member that scored best in the same pool is 0.63 points per episode better than the exported one, and the exported one is no better than the pool’s median member (-1.92 against -1.78). On the episodes that selected it, the best member looks 0.90 better: the best of 128 noisy scores is inflated, by 0.27 here, a winner’s curse that an earlier version of this analysis did not correct; every comparison with the best member below is re-scored on held-out episodes. At the last snapshot the exported champion averages -0.15 while 67 of its peers score above parity. The reported curve also loses competence that the population does not: summed over the control runs, the exported champion’s held-out score fell by 7.8 points between consecutive snapshots, the best member’s by 2.0 and the median member’s by 1.9, and the exported champion fell further in 6 of 6 runs. The population rarely gets worse; the individual we report does. The counter cannot do better: it is inherited on every replacement, so it records how long a lineage has survived, not how good its current member is.

The rule can be replaced without external information. In a post hoc re-scoring of the same snapshots, ranking the pool by an internal round robin of 4,096 games per snapshot, 0.8% of the training budget, raised the exported member’s mean score by $+0.52$ (Appendix, Table 6). Its correlation with true skill saturates near $+0.57$: internal fitness measures skill against the current population, which is not the same as skill against an unseen opponent, so held-out evaluation remains necessary for selection as well as for reporting.

That analysis chose its budgets after seeing the data. A preregistered test on 12 fresh control runs (repository; Appendix A.10) confirmed the direction at a smaller size. Exporting the winner of an internal round robin of 1,024 games per snapshot instead of the longest streak raised the exported member’s held-out score from -2.33 to -2.16 (higher in 9 of 12 runs, one-sided $p = 0.010$; Figure 4), a gain of $+0.17$ against $+0.46$ for the same budget post hoc, and closed 43% of the gap to the best member re-scored on held-out episodes. It did not detectably reduce the curve’s losses of competence (summed declines 1.37 against 1.98, $p = 0.16$).

3.5 The mutation step is not the cause

A third explanation for the swings is a mutation step too large for the landscape (Appendix A.3). Halving it changes nothing that matters: 3/3 runs learned, 26% of checkpoints are above parity (control 26%), and the volatility of the runs that learned matches the control’s ($\delta = +0.00$, $p = 1.000$). Doubling it costs runs (2/3 learned) but not stability ($\delta = -0.17$, $p = 0.857$ among the runs that learned). Over all runs the larger step even looks calmest (volatility 0.38 against 0.67), because a run that never learns contributes a flat line, which is why stability is only compared among runs that learned. A population of 32, run once, never learned (0/1).

3.6 An archive of past champions: as a parent, and as a test

The standard remedy for forgetting is a hall of fame (Rosin and Belew, 1997). We tested two readings of it. Our first implementation applied Ha’s replacement rule to archive games, so a winning archived genome became the parent of the member it beat: an archive used as a *parent*. With $p = 0.25$ that copies an old genotype back into the pool in 12.0% of all games, and it abolishes

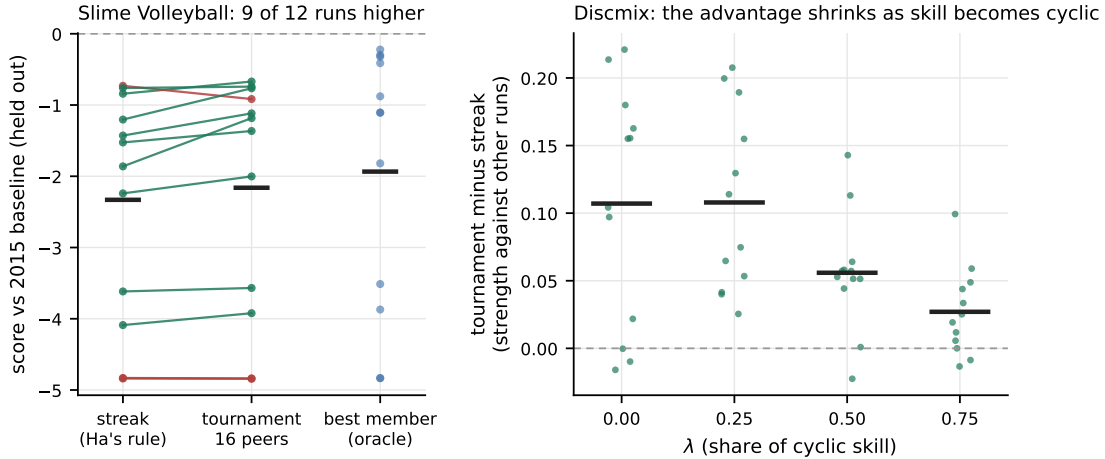


Figure 4: The preregistered export-rule test, both rules applied to the same stored populations of fresh runs. Left: Slime Volleyball, each run’s mean held-out score of the exported member under the streak rule and under a 16-peer internal tournament (green: higher, red: lower), and the pool’s best member re-scored held out; bars are means. Right: discmix, each run’s advantage of the tournament in strength against other runs’ populations, by the share λ of cyclic skill; bars are means.

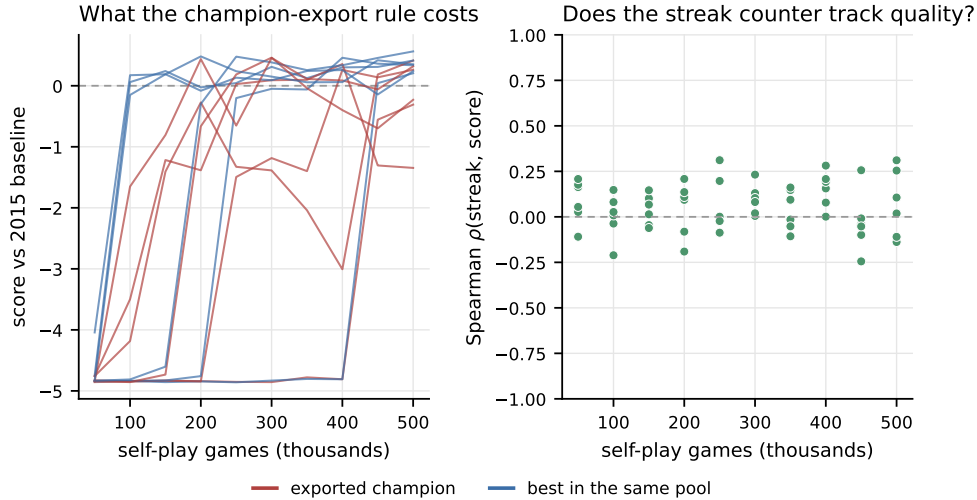


Figure 5: Left: the exported champion (red) and the best member of the same population (blue), both re-scored on held-out episodes, every control seed. Right: Spearman correlation between the streak counter and score against the 2015 policy, every seed and snapshot.

learning: 1/6 runs learned to rally, against 6/6 for the control, and 0/4 at a higher dose or with a full-run archive (Table 2). These runs are also the most “stable” in the matrix, with near-zero volatility and drawdown, which is why stability is only compared among runs that learned.

The literature’s reading uses the archive as a *test*: archived champions are opponents, never parents. In *hof-eval-v2* an archive game that the population member loses costs it its slot, but the replacement comes from the living pool; the archive holds a champion from every 1,000 games of the run (capacity 512). Done this way the archive is neither destructive nor helpful (Figure 6). 5/6 runs learned to rally and 4/6 produced an above-parity champion, against 6/6 and 6/6 for the

control, and at 6 runs a side no outcome is distinguishable from the control: checkpoints above parity 18% against 26% ($\delta = -0.33$, $p = 0.370$), mean level over the last 100,000 games $\delta = -0.22$ ($p = 0.589$), best held-out champion $\delta = +0.00$ ($p = 1.000$). The archive’s own win rate shows why it has little to do. Early in a run, archived champions win 0.46–0.50 of their games; in every run whose population learned they end at 0.11–0.16, so 21–22% of late games produce no selection event (an archive game the member wins overwrites nothing). Skill is transitive (§3.3), so a past champion is simply a weaker player. In the run that never learned, the archive’s win rate stayed at 0.50. Logging that win rate is a cheap check on whether an archive still has anything to teach; here it tracked progress, but it is a description of 6 runs, not a validated predictor, and in the replication it did not hold (§3.8). An earlier version of this condition credited a win to the living peer that supplied the replacement genes, although that peer had not played; since the longest streak decides what enters the archive, the bug changed the whole trajectory. Those runs are superseded and excluded. With the bug the archive looked measurably worse than the control and learned later; the corrected runs do not, and the decision log shipped with the data gives both side by side.

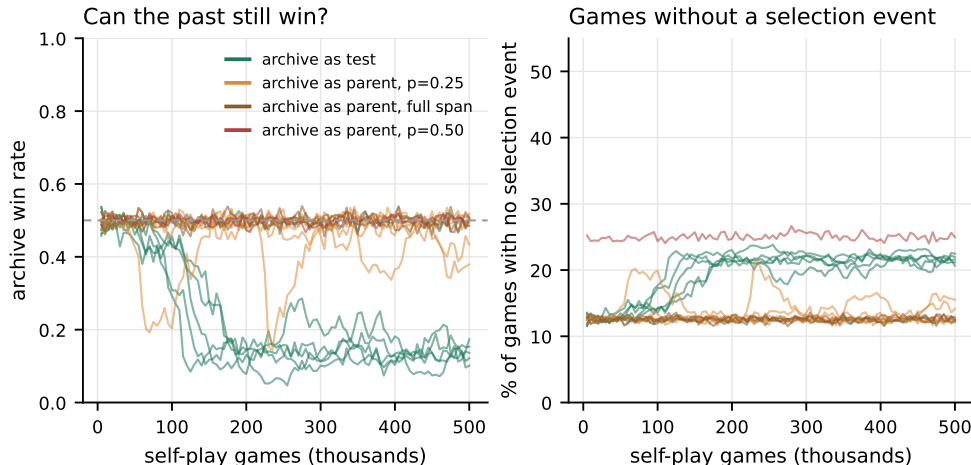


Figure 6: Left: the archive’s win rate against the current population, every archive run. Right: the share of games that produce no selection event, because an archive game the population member wins overwrites nothing.

3.7 Algorithm families reach a similar ceiling

Two further families change the machinery with policy and environment fixed: a generational GA in the style of Ha’s 2015 experiment, which ranks by an explicitly computed fitness and uses crossover, and a self-play evolution strategy (Salimans et al., 2017; Ha, 2017), which reports its distribution mean and so has no export rule at all (Figure 7). With the archive-as-test condition as a fourth family, all four reach a similar ceiling: the single highest end-of-run champion of the matrix, +0.50, is a self-play ES seed, above every control seed (best +0.41). In their six seeds they seemed to differ in their floor: only the control learned to rally and produced an above-parity champion in every seed (6/6, 6/6); the generational GA managed 5/6 and 5/6, the ES 4/6 and 2/6, the archive as test 5/6 and 4/6, and the spread of end-of-run champions was 3.0–3.5 times the control’s for the other three families. That difference did not survive fresh seeds. In the replication (§3.8) twelve new control runs learned in 10/12, exactly as often as the archive as test on the same seeds (10/12), and their end-of-run spread, 1.92, was as wide as the other families’ in the matrix (2.00–2.29; the

control’s own six seeds: 0.66). The generational GA and the ES were not rerun, so their reliability is unknown beyond six seeds. What survives is methodological: a study with one seed per family could have crowned the ES on the strength of a seed that happened to work, and six seeds per family produced a reliability ranking that twelve more control runs did not support. The cross-run tournament, in which the final champions of all 41 single-population runs play each other with no external opponent, shows the same pattern in these runs (Appendix A.5): every control champion rates above zero (worst +225), while each of the other three families has a champion at or below −576; across runs, too, skill is transitive (0.1% of decided triads cyclic). The generational GA ranks by a computed fitness, which §3.4 suggests should help, and still does not come out ahead; crossover between weight vectors and its coarser, generational update are plausible reasons we cannot separate here. The self-play ES was tuned only at pilot scale, so its result is a statement about the configuration we could afford, not about the method.

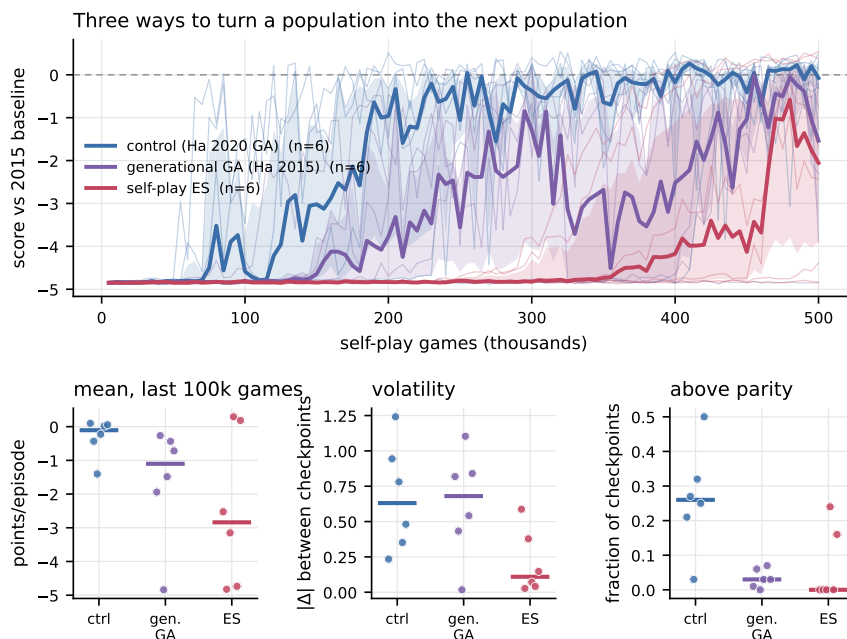


Figure 7: Three families at an identical budget. Top: median and inter-quartile band. Bottom: per-seed level and volatility of the last 100,000 games and the share of checkpoints above parity (ctrl: control; gen. GA: generational GA).

3.8 A preregistered replication

The results above rest on six seeds per main condition, and the design was revised while it ran (Section 6). The claims about Ha’s GA and the archive (Sections 3.1–3.4 and 3.6) were therefore tested again on fresh seeds, with every test, threshold and decision rule committed before the first run: 12 new runs each of the control, the archive as parent and the archive as test, 36 in all, measured with the code that produced the results above and analysed once. 6 of 6 claims replicated (Table 8 in Appendix A.6). The internal transition preceded parity in 10/10 eligible runs; checkpoint skill was transitive ($\rho = +0.79$, 0.5% cyclic triads); the exported individual ranked on average 56 of 128 in its pool, with $\rho(\text{streak}, \text{skill}) = +0.04$, and lost 21.2 points between snapshots where the pool’s best member lost 7.6; the archive as parent learned in 1/12 runs; the archive as test was again indistinguishable from the control (every $p \geq 0.44$).

Three statements come out weaker than six seeds suggested. Not every control run learns: 10/12 did, which meets the preregistered threshold but not “every seed” (and undoes the control’s apparent edge in reliability, §3.7). A falling archive win rate does not by itself indicate learning: it fell to 0.06–0.16 in the archive-as-test runs that learned, but also to 0.21 in one that never did. And the declines comparison, as preregistered, scored the pool’s best member on the episodes that chose it (§3.4); re-scored on held-out episodes the exported champion still lost more, 19.5 points against 7.7, but in 9 of 12 runs (one-sided sign test $p = 0.073$), short of the threshold the preregistered measure met; the verdict stands as preregistered, and the held-out version is the more conservative reading. The comparison of algorithm families (§3.7) and the unequal-power block (§4) were not part of the replication; the export-rule test of §3.4 was preregistered on fresh runs of its own.

3.9 Beyond one game, one archive and one topology

Three questions the main design could not answer were run afterwards as preregistered experiments in the repository’s lab (`docs/lab.md`). Its GA reproduces the study’s kernels bit for bit, and its NEAT, given a genome that encodes the study’s network, plays bit-identical games; the tables are in Appendix A.11.

Cyclic skill. Slime Volleyball turned out almost purely transitive, which could make “the population is not cycling” a property of the game and the export-rule failure an artefact of transitivity. In a synthetic game whose mix of transitive and cyclic skill is set by one parameter λ , with exact expected scores (96 runs), cycling rose as designed (13.5% cyclic champion triads at $\lambda = 0.75$), yet the streak rule exported an individual ranked on average 52–63 of 128 in its own pool at every λ , its streak barely related to its true strength ($\rho = +0.07$ to $+0.13$). The internal tournament of §3.4 helped there too, judged against other runs’ populations (better in 42 of 48 runs), but its advantage shrank as skill became more cyclic, from $+0.11$ to $+0.03$ ($\rho = -0.43$, $p = 0.001$). An archive used as a test did not help more as skill became cyclic (trend $p = 0.116$).

Another archive. The archives of §3.6 hold past champions in time order. One organised by behaviour instead, one champion per cell of a MAP-Elites-style grid over a behaviour descriptor and used as a test (60 runs), had no detectable effect in Slime Volleyball (all $p \geq 0.29$ against the control) although it filled 37–53 of 64 cells; it did not help more as skill became cyclic ($p = 0.072$), and where skill was most cyclic it did not beat the time-ordered archive ($p = 0.34$). Neither archive design helped.

Another topology. NEAT (Stanley and Miikkulainen, 2002), evolving structure under the generational GA’s evaluation and the same budget, never learned to rally (0 of 12 runs; final champions -4.84), detectably worse than the generational GA (Cliff’s $\delta = -0.92$, $p = 0.0008$), while growing 23–39 hidden nodes. Exploratory runs (not preregistered) located the failure in its reproduction machinery as configured: started from the study’s network it still failed (0 of 4 learned), while its loop with speciation and structural mutation removed learned in 1 of 4. This is one configuration with mostly published settings (Stanley and Miikkulainen, 2004), untuned for this game.

4 Unequal power (exploratory)

This block was designed after the main matrix had run and re-implemented twice before the runs reported here; it is exploratory and we draw no confirmatory conclusion from it. Two populations of 128 play only each other, a share 0.25 of each side’s games crossing over; in two conditions one side has a 12–16–16–3 network with 531 parameters against 273, in one with a common per-parameter σ and in one with the larger side’s σ scaled to equalise the mutation step norm. A symmetric two-population control has 273 a side (Appendix, Figure 12 and Table 11).

Of 18 runs, 1 ended with both pools holding a member above parity and 13 in runaway dominance, 4 of them in the symmetric control, where the sides differ only in their seed; neither capacity condition is distinguishable from that control (Appendix A.9). The pattern of one side collapsing is consistent with disengagement (Cartlidge and Bullock, 2004).

5 Discussion

Three sources of variance, not two. Coevolutionary methodology separates the *ecology*, which selects, from the *yardstick*, which measures. This study found a third component with its own error term: the *promotion rule* that names one artefact out of the population. In the original control runs the exported individual lost about 3.8 times as much ground between population snapshots as the population’s best member did, both re-scored on held-out episodes, and a champion curve hides this, because it looks the same whether its swings come from the population or from the choice of individual.

A hypothesis about other evolutionary loops. The failure does not depend on skill being transitive: with cyclic skill the streak rule still exported a middling individual, and the internal tournament’s advantage shrank as skill became more cyclic (§3.9). Measuring candidates against their own population is a weaker guide to quality when quality is relative. Any loop that promotes one artefact out of a population has a promotion rule, and that rule is usually cheaper than a measurement of quality. LLM-driven program evolution (Novikov et al., 2025; Lange et al., 2025) promotes one program per generation or per run on the strength of an evaluator that may be noisy, small or proxy-based. Gideoni et al. (2026) report the closely related failure in that setting: agentic scaffolds selected on small validation sets lose much of their apparent advantage on held-out data, because the selection was partly by chance. We do not test program evolution here. We state as a hypothesis that its reported progress curves contain a promotion-rule component of the kind measured above, and propose the same diagnostic: before attributing a noisy progress curve to the search, check whether the statistic used to promote correlates with held-out quality across the candidates it chose between, and compare the promoted candidate with the population’s median. Both comparisons need data that played no part in the choice: the best of many noisy candidates looks better on the data that chose it, which in this study inflated the best member’s score by 0.27 points per episode (§3.4).

Diagnose before applying a remedy. A cheap within-run round robin found skill transitive and so predicted that an archive, the remedy for intransitivity, would have little to do; used as a test it changed nothing measurable, used as a parent it pulled the loop backwards.

6 Limitations

- **One environment**, symmetric, zero-sum and fully observed: the most favourable setting for purely relative selection, and one whose skill structure is largely transitive. The synthetic game of §3.9 makes skill cyclic but is a second setting, not a second physical game; a physical game with cyclic skill remains untested.
- **Few seeds**. 6 runs per condition separate large effects from seed noise, not small ones; several conditions have fewer. Secondary comparisons are uncorrected for multiplicity. The claims about Ha’s GA and the archive were confirmed on 12 fresh runs per condition (§3.8) and the export-rule remedy was preregistered on fresh runs; the family comparison and the unequal-power block were not replicated. Six seeds were demonstrably too few once: they suggested a reliability ranking of the families that the replication’s fresh control runs did not reproduce (§3.7).
- **Post hoc design changes**. The design was revised while it ran (seed counts, the archive conditions, the replacement of the population sweep by the unequal-power block, the rerun of the archive-as-test condition after a bug fix). Each change is logged with what was known when it was made, but the final design is not the one pre-registered; the replication of §3.8 was.
- **Scope of the measurements**. The export analysis is control-only, because only the control stores population snapshots, and the ES was tuned only at pilot scale. The archive as a parent has one design, as a test two (§3.6, §3.9); a curated archive, or one that picks opponents by what they teach, is untested. Of export rules only the streak rule and a 16-peer tournament were tested.
- **Topology**. Every result outside §3.9 comes from families that evolve one fixed network. One topology-evolving algorithm was run, in one configuration (§3.9); a NEAT tuned for self-play is untested.
- **The unequal-power block is exploratory** and underpowered for its most interesting comparison.

7 Reproducibility

The code, every run file, the protocol, the decision log and this paper’s sources are in the repository. A 500,000-game run would take about 13 core-hours on the reference implementation; the compiled port runs 26 times faster on one core. It is validated by driving both implementations from one stream of serve velocities: 200 paired games agree on every ball and agent position, every rally end, every score and every episode length, over 304,812 environment steps (Appendix, Table 22). Every table and every number in this paper is generated from the analysis files, and every analysis file records the hash of each raw file it was computed from; a consistency check, run on every push, fails if any number does not follow from the files on disk. Rerunning the analysis pipeline reproduces the committed analysis files.

Acknowledgements

The environment, the 2015 baseline policy and the genetic algorithm studied here are David Ha’s (Ha, 2015, 2020), released under the Apache-2.0 licence; the compiled port in this study transcribes

them. For background on neuroevolution and coevolution see Risi et al. (2026).

References

- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- David Balduzzi, Marta Garnelo, Yoram Bachrach, Wojciech M. Czarnecki, Julien Pérolat, Max Jaderberg, and Thore Graepel. Open-ended learning in symmetric zero-sum games. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 434–443, 2019.
- Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3–4):324–345, 1952. URL <https://academic.oup.com/biomet/article-abstract/39/3-4/324/326091>.
- John Cartlidge and Seth Bullock. Combating coevolutionary disengagement by reducing parasite virulence. *Evolutionary Computation*, 12(2):193–222, 2004. URL <https://ieeexplore.ieee.org/document/6790490>.
- Dave Cliff and Geoffrey F. Miller. Tracking the red queen: Measurements of adaptive progress in co-evolutionary simulations. In *Advances in Artificial Life (ECAL 1995)*, volume 929 of *Lecture Notes in Computer Science*, pages 200–218. Springer, 1995. doi: 10.1007/3-540-59496-5_300.
- Wojciech M. Czarnecki, Gauthier Gidel, Brendan Tracey, Karl Tuyls, Shayegan Omidshafiei, David Balduzzi, and Max Jaderberg. Real world games look like spinning tops. In *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020.
- Sevan G. Ficici and Jordan B. Pollack. Pareto optimality in coevolutionary learning. In *Advances in Artificial Life (ECAL 2001)*, volume 2159 of *Lecture Notes in Computer Science*, pages 316–325. Springer, 2001. doi: 10.1007/3-540-44811-X_34.
- Sevan G. Ficici and Jordan B. Pollack. A game-theoretic memory mechanism for coevolution. In *Genetic and Evolutionary Computation (GECCO 2003)*, volume 2723 of *Lecture Notes in Computer Science*, pages 286–297. Springer, 2003. doi: 10.1007/3-540-45105-6_35.
- Yonatan Gideoni, Sebastian Risi, and Yarin Gal. Simple baselines are competitive with code evolution, 2026.
- David Ha. Neural slime volleyball. Blog post, <https://blog.otoro.net/2015/03/28/neural-slime-volleyball/>, 2015.
- David Ha. ESTool: Evolution strategies tool. <https://github.com/hardmaru/estool>, 2017.
- David Ha. Slime volleyball gym environment. <https://github.com/hardmaru/slimevolleygym>, 2020.
- Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. doi: 10.1609/aaai.v32i1.11694.

- Robert Tjarko Lange, Yuki Imajuku, and Edoardo Cetin. ShinkaEvolve: Towards open-ended and sample-efficient program evolution, 2025.
- Henry B. Mann and Donald R. Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1):50–60, 1947. doi: 10.1214/aoms/1177730491.
- Thomas Miconi. Why coevolution doesn’t “work”: Superiority and progress in coevolution. In *Genetic Programming (EuroGP 2009)*, volume 5481 of *Lecture Notes in Computer Science*, pages 49–60. Springer, 2009. doi: 10.1007/978-3-642-01181-8_5.
- Stefano Nolfi and Paolo Pagliuca. Global progress in competitive co-evolution: A systematic comparison of alternative methods. *Frontiers in Robotics and AI*, 2025. doi: 10.3389/frobt.2024.1470886.
- Alexander Novikov et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery, 2025.
- Elena Popovici, Anthony Bucci, R. Paul Wiegand, and Edwin D. de Jong. Coevolutionary principles. In *Handbook of Natural Computing*, pages 987–1033. Springer, 2012. doi: 10.1007/978-3-540-92910-9_31.
- Sebastian Risi, Yujin Tang, David Ha, and Risto Miikkulainen. *Neuroevolution: Harnessing Creativity in AI Agent Design*. MIT Press, 2026. ISBN 9780262054768. Open-access online edition at <https://neuroevolutionbook.com>.
- Christopher D. Rosin and Richard K. Belew. New methods for competitive coevolution. *Evolutionary Computation*, 5(1):1–29, 1997. URL <https://direct.mit.edu/evco/article-abstract/5/1/1/790/New-Methods-for-Competitive-Coevolution>.
- Tim Salimans, Jonathan Ho, Xi Chen, Szymon Sidor, and Ilya Sutskever. Evolution strategies as a scalable alternative to reinforcement learning, 2017.
- Deacon Seals and Daniel R. Tauritz. Tripping over the past: Measuring deceptive progress in competitive coevolutionary algorithms employing hall of fame. In *Applications of Evolutionary Computation (EvoApplications 2026)*, Lecture Notes in Computer Science. Springer, 2026. doi: 10.1007/978-3-032-23607-4_32.
- Luca Simione and Stefano Nolfi. Long-term progress and behavior complexification in competitive coevolution. *Artificial Life*, 26(4):409–430, 2020. URL <https://direct.mit.edu/artl/article/26/4/409/97302>.
- Kenneth O. Stanley and Risto Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2):99–127, 2002. URL <https://direct.mit.edu/evco/article/10/2/99/1123/Evolving-Neural-Networks-through-Augmenting>.
- Kenneth O. Stanley and Risto Miikkulainen. Competitive coevolution through evolutionary complexification. *Journal of Artificial Intelligence Research*, 21:63–100, 2004.

A Supplementary results

A.1 The reference run in detail

The best checkpoint of the reference run, at 439,000 games, wins 42%, draws 52% and loses 6% of held-out episodes; the final one wins 19%, draws 64% and loses 17% (Table 3). On the sweep, 112/500 of its checkpoints score above parity. What transfers to the 2015 policy is a general ability not to lose, not a counter-strategy: the population never sees that opponent.

The single-run version of this study reported that the swings shrink as the level rises. With 6 seeds that is testable, and it does not hold (Table 4): from the first to the last 100,000-game window the control’s mean rises from -4.32 to -0.32 and the checkpoints above parity from 1/120 to 64/120, but the within-run spread of checkpoint scores is 0.74 in the first window and 0.72 in the last.

Table 3: The reference run: final and best checkpoint, held out.

checkpoint	games	score vs 2015 baseline	won / drawn / lost	mean rally
final	500,000	$+0.036 \pm 0.734$ (s.e.m. 0.023)	19% / 64% / 17%	3000 steps
peak	439,000	$+0.496 \pm 0.856$ (s.e.m. 0.027)	42% / 52% / 6%	3000 steps
Ha (2020), same algorithm and budget	500,000	$+0.353 \pm 0.728$	—	—

Table 4: The control in 100,000-game windows: mean score across seeds, within-run spread of checkpoint scores, spread of window means across seeds, and checkpoints above parity.

games	mean across seeds	within-run s.d.	spread across seeds	checkpoints above parity
0–100,000	-4.32	0.74	0.57	1/120
100,000–200,000	-2.76	0.83	1.78	14/120
200,000–300,000	-1.46	0.69	1.75	34/120
300,000–400,000	-1.01	0.51	1.71	45/120
400,000–500,000	-0.32	0.72	0.52	64/120

A.2 The export rule over training, and promotion rules

Table 5 follows the comparison of §3.4 through training: at every snapshot the exported individual sits in the middle of its population, the streak counter’s correlation with skill stays near zero, and the gap to the best member of the pool persists. Table 6 and Figure 8 replace the streak rule, post hoc on the same snapshots, with an internal round robin of increasing size; the rule uses no information the algorithm does not already have, so it is deployable, unlike the best-in-pool oracle. The prospective test of the 16-peer rule is Appendix A.10.

Table 5: The exported champion against its own population, original control runs, averaged across seeds at each snapshot. The best member is chosen on 30 episodes per member; both members’ scores and the gap are re-scored on 1,000 held-out episodes, because on the episodes that chose it the best member’s score is inflated. Rank, ρ and the parity count use the 30-episode scores.

games	exported champion	best in the same pool	gap	exported rank	$\rho(\text{streak, score})$	above parity in pool	mean pairwise genotype distance
50,000	-4.81	-4.69	0.12	50 / 128	+0.09	0	12.66
100,000	-3.98	-2.40	1.58	71 / 128	+0.00	1	10.79
150,000	-2.97	-2.28	0.69	58 / 128	+0.04	5	10.43
200,000	-1.93	-1.59	0.34	54 / 128	+0.05	13	10.13
250,000	-1.35	-0.69	0.66	50 / 128	+0.07	23	9.07
300,000	-1.07	-0.66	0.41	44 / 128	+0.10	32	10.50
350,000	-1.34	-0.70	0.64	73 / 128	+0.04	34	10.53
400,000	-1.25	-0.57	0.69	56 / 128	+0.15	47	12.61
450,000	-0.38	+0.24	0.62	76 / 128	-0.03	49	12.23
500,000	-0.15	+0.35	0.50	61 / 128	+0.07	67	11.38

Table 6: Promotion rules on the original control runs’ population snapshots, post hoc. “Gap closed”: share of the distance between the streak-exported and the best member recovered by the rule. The oracle row is scored on the episodes that chose it and is therefore inflated, so the gap is overstated and the share closed understated; the rules’ own rows are unbiased.

promotion rule	games spent ranking	mean score of the exported individual	volatility of the reported series	gap to best closed	$\rho(\text{rule statistic, true skill})$
winning streak (Ha’s rule)	0	-1.95	0.84	—	+0.04
<i>pick the population median</i>	0	-1.83	0.57	—	—
internal round robin, 4 peers each	256	-1.77	0.72	22%	+0.32
internal round robin, 8 peers each	512	-1.55	0.77	50%	+0.41
internal round robin, 16 peers each	1,024	-1.49	0.70	57%	+0.49
internal round robin, 32 peers each	2,048	-1.47	0.73	60%	+0.53
internal round robin, 64 peers each	4,096	-1.43	0.68	65%	+0.57
<i>best in pool (oracle, not deployable)</i>	—	-1.14	0.61	100%	+1.00

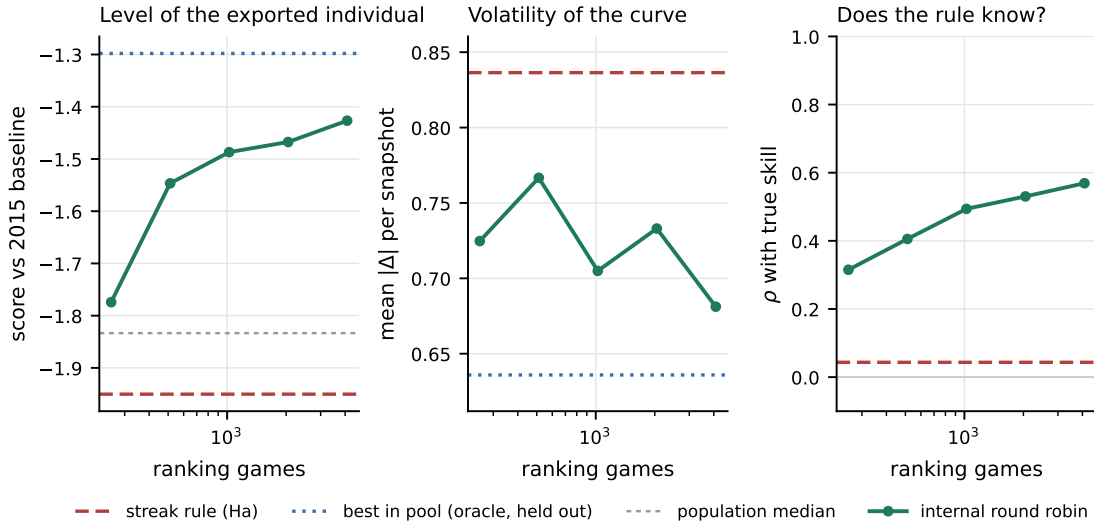


Figure 8: Promotion rules as a function of the games spent ranking (post hoc, original control runs): level of the exported individual, volatility of the reported curve, and rank correlation between the rule’s statistic and true skill. The oracle lines show the best member re-scored on held-out episodes (Table 5).

A.3 Mutation scale and population size

Figure 9 shows the sweeps discussed in §3.5. The population sweep was cut to a single run of the smaller population when the unequal-power block replaced it; the larger population was never run.

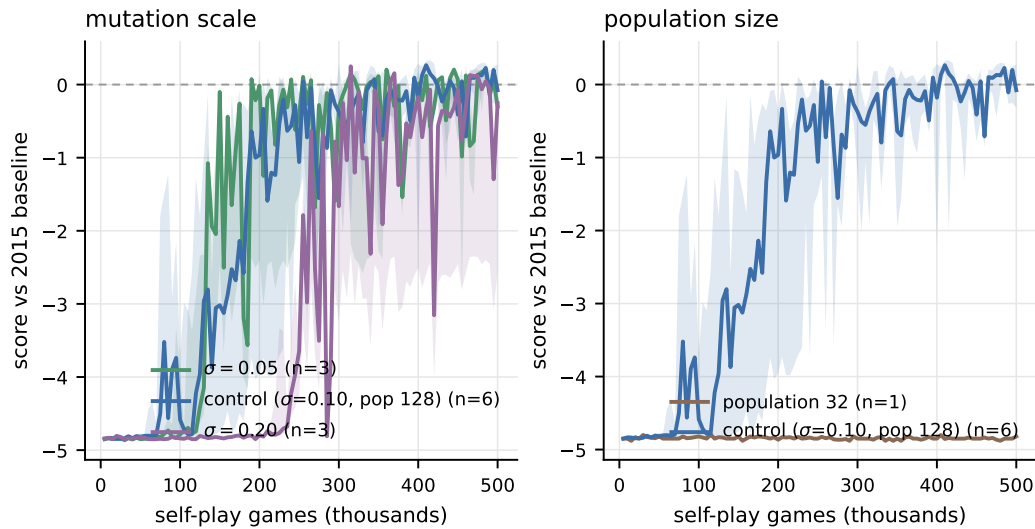


Figure 9: Mutation scale (left) and population size (right) against the control, median and inter-quartile band across seeds.

A.4 The archive conditions

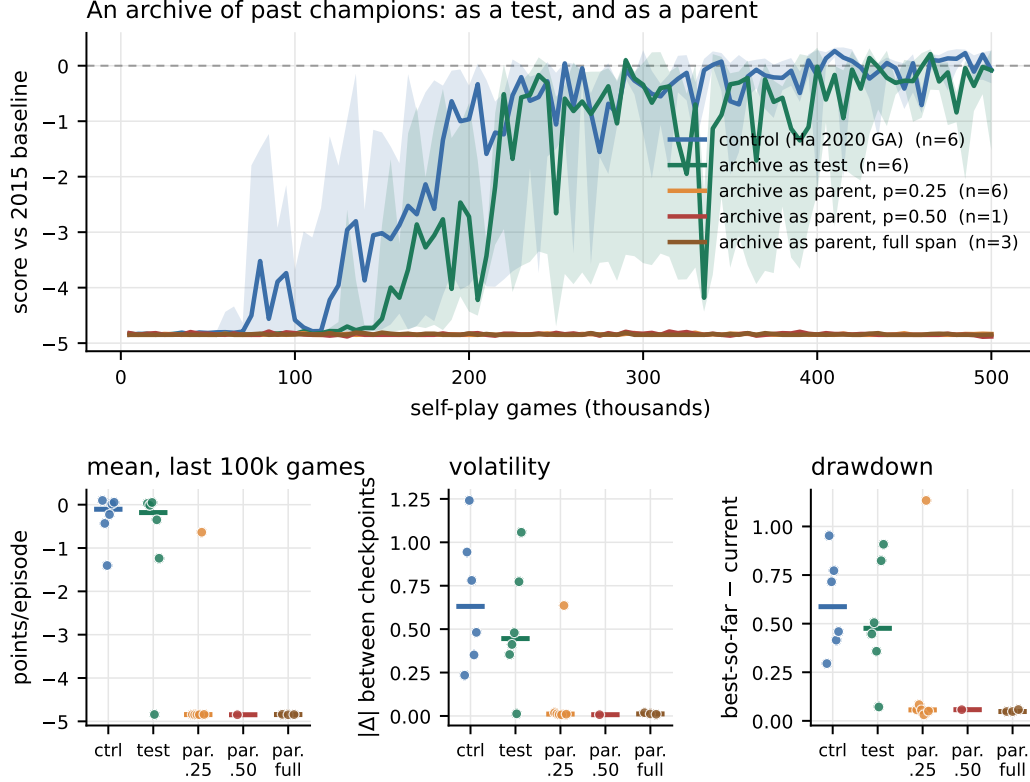


Figure 10: The archive conditions against the control. Top: median and inter-quartile band. Bottom: per-seed level, volatility and drawdown of the last 100,000 games (ctrl, control; test, archive as test; par., archive as parent at the dose or span given in the legend).

A.5 The cross-run tournament

Scores against a single frozen opponent can reward specialising against it. The final champion of every single-population run therefore played the final champion of every other (Figure 11, Table 7). The control’s median rating is +295 and its worst champion +225; the archive as test has a higher median (+381) but also a champion at −596; the generational GA and the ES have medians of +56 and +99. The highest median belongs to the smaller mutation step (+472), a side condition with fewer runs.

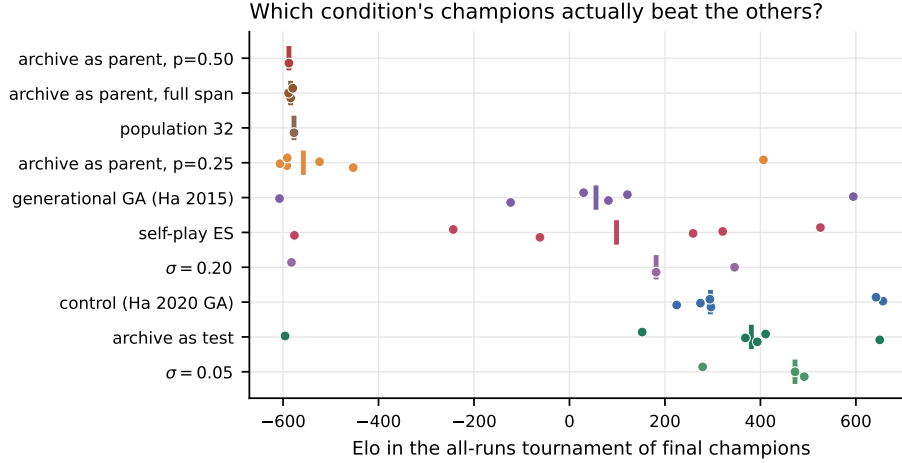


Figure 11: Bradley–Terry ratings on the Elo scale of every run’s final champion in one all-play-all tournament; bars mark condition medians.

Table 7: Cross-run tournament of final champions, by condition.

condition	runs	median Elo	best run	worst run
control (Ha 2020 GA)	6	+295	+657	+225
archive as test, full span	6	+381	+650	−596
archive as parent, $p=0.25$	6	−557	+406	−606
archive as parent, $p=0.50$	1	−588	−588	−588
archive as parent, full span	3	−584	−580	−589
generational GA (Ha 2015)	6	+56	+595	−607
self-play ES	6	+99	+526	−576
$\sigma = 0.05$	3	+472	+492	+279
$\sigma = 0.20$	3	+181	+346	−582
population 32	1	−577	−577	−577

A.6 The preregistered replication

The design, hypotheses, tests and decision rules are in `results/replication/PREREGISTRATION.md`, committed before the first replication run; `replication.py` is the analysis, run once. Its measurements are the main text’s own and, applied to the original seeds, reproduce the committed analysis files bit for bit. One run was trained in the session’s container and the rest on GitHub’s runners, after the container was reclaimed twice mid-run; a run repeated on both machines gave identical files (`results/matrix/decisions.md`).

Table 8: The preregistered replication: 12 fresh runs per condition, every criterion fixed before the first run. Tests in the Holm family hold when rejected at family-wise $\alpha = 0.05$.

claim	criterion	result	holds
C1	internal transition before parity ($\text{lag} > 0$)	10/10 runs, $p = 9.8\text{e-}04$	yes
C2	control runs that learn to rally	10/12	yes
C2	latest / earliest internal transition ≥ 3	5.8×	yes
C3	$\rho(\text{Elo}, \text{time}) > 0$ within runs	12/12 runs (mean $+0.79$), $p = 2.4\text{e-}04$	yes
C3	cyclic share of decided triads $< 1\%$	4/849 (0.5%)	yes
C4	exported individual outside its pool's top quarter	12/12 runs (mean rank 56), $p = 2.4\text{e-}04$	yes
C4	$\rho(\text{streak}, \text{skill})$ inside ± 0.2 (90% CI)	$+0.04 [+0.02, +0.06]$	yes
C4	exported declines more than the best member	10/12 runs (21.2 vs 7.6), $p = 0.019$	yes
C5a	archive as parent learns less often than control	1/12 vs 10/12, $p = 3.2\text{e-}04$	yes
C5b	archive as test vs control: all four $p \geq 0.05$	final $\delta +0.01$ p 0.98, peak $\delta -0.19$ p 0.44, above $\delta -0.15$ p 0.56, late $\delta -0.08$ p 0.76	yes

A.7 Transitivity at 5,000-game spacing

Table 9 repeats the within-run tournament of Section 3.3 on every stored champion of the control runs, one per 5,000 games, with two rules for when a pair counts as decided. Under the main text’s deadband rule the pooled cyclic share is 1.0%, and 4.5% among triads spanning at most 50,000 games; under an exact sign test it is 0.01%. The difference is sampling noise: with 20 games per pair, the margin between two near-equal policies has a standard error of the order of the deadband. The same tournament shows the exported champion moving at this resolution while the population does not: among adjacent champions whose difference the sign test decides, the later one wins 88/144 times (61%).

Table 9: Within-run tournament at 5,000-game spacing, control runs.

run	$\rho(\text{Elo}, \text{time})$	cyclic, ± 0.25 rule	cyclic, sign test	within 50k games, sign test	next beats previous
control_s101	+0.95	0.69% (774/112,190)	0.00% (0/51,268)	0.00% (0/272)	10/14
control_s102	+0.76	0.55% (541/98,864)	0.02% (11/51,840)	0.23% (1/430)	18/35
control_s103	+0.81	3.82% (3,675/96,104)	0.00% (1/21,569)	0.00% (0/198)	7/13
control_s104	+0.68	0.50% (431/86,322)	0.04% (16/43,215)	0.23% (1/440)	21/34
control_s105	+0.91	0.22% (241/108,654)	0.00% (1/58,693)	0.00% (0/454)	16/24
control_s106	+0.85	0.30% (253/85,284)	0.01% (6/46,791)	0.86% (3/349)	16/24
<i>all control runs</i>	—	1.01% (5,915/587,418)	0.01% (35/273,376)	0.23% (5/2,143)	88/144

A.8 Stronger external opponents

The 2015 baseline never sees its opponent, so “above parity” against it is a low bar. Ha (2020) also publishes two trained policies: a self-play GA champion of the same 273-parameter class, and a 743-parameter CMA-ES policy trained against the baseline, which score $+0.35$ and $+1.15$ against it. Every final champion played 200 games against each, half on each side, on the compiled environment; for networks of both sizes it reproduces the reference environment bit for bit (40 games, 120,000 steps compared).

The baseline ranks the final champions much as the stronger opponents do (Spearman $\rho = +0.87$ against the zoo GA and $+0.89$ against the zoo CMA-ES, over 41 champions), so the comparisons in the main text do not hinge on it. It does flatter them: 0 of the 41 beat the published GA champion (the best scores -0.04), and of the 10 above parity against the baseline, 5 also beat the CMA-ES policy (Table 10). The published GA champion is a single export its author chose to release, not an unselected endpoint. The CMA-ES policy, trained only against the baseline, is the stronger of the two by the baseline’s measure yet loses head to head to the GA champion ($+0.30$ for the GA): specialising against the yardstick, in one comparison.

Table 10: Final champions against the 2015 baseline (held out) and the two slimevolleygym zoo policies, points per episode, by condition.

condition	runs	vs 2015 baseline	vs zoo GA	vs zoo CMA-ES	beat zoo GA	beat zoo CMA-ES
control (Ha 2020 GA)	6	-0.15	-0.98	-0.10	0/6	2/6
archive as test, full span	6	-1.01	-1.72	-1.25	0/6	1/6
archive as parent, p=0.25	6	-4.10	-4.34	-3.95	0/6	1/6
archive as parent, p=0.50	1	-4.84	-4.99	-4.87	0/1	0/1
archive as parent, full span	3	-4.84	-4.98	-4.92	0/3	0/3
generational GA (Ha 2015)	6	-2.00	-3.83	-1.47	0/6	1/6
self-play ES	6	-2.08	-3.34	-2.41	0/6	1/6
sigma = 0.05	3	-0.17	-0.62	+0.10	0/3	2/3
sigma = 0.20	3	-1.66	-2.79	-2.08	0/3	0/3
population 32	1	-4.86	-4.99	-4.88	0/1	0/1

A.9 Unequal power (exploratory)

Of 18 runs, 1 ended with both pools holding a member above parity; 10 ended with exactly one, 7 with neither. 13 ended in runaway dominance (one side winning more than a share 0.9 of the cross-population games at the end), 4 of them in the symmetric control, where the sides differ only in their seed. Neither capacity condition is distinguishable from that control (common σ : $\delta = -0.33$, $p = 0.394$; matched step: $\delta = +0.11$, $p = 0.818$). The larger side led in 2/6 runs at common σ and in 5/6 with the step norm matched ($\delta = +0.67$, $p = 0.065$), which suggests, but at 6 seeds a side does not establish, that extra capacity converts into advantage only when the mutation scale is adjusted to it. The pattern of one side collapsing is consistent with disengagement (Cartlidge and Bullock, 2004).

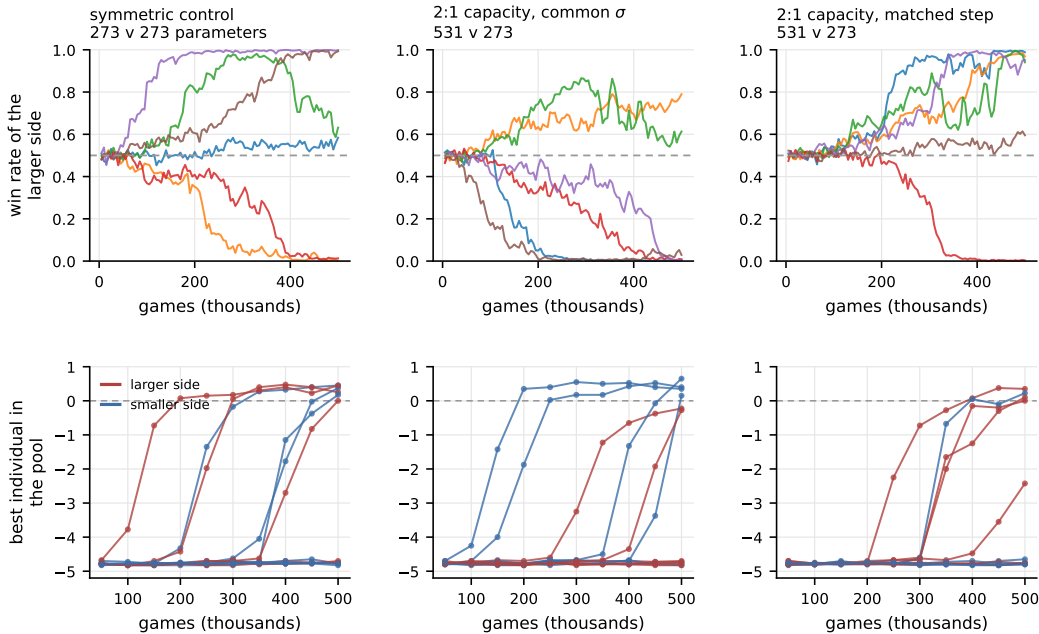


Figure 12: Two-population runs (exploratory, Section 4). Top: the larger side’s win rate in cross-population games, one line per seed. Bottom: the best member of each pool against the 2015 policy (red: larger side, blue: smaller).

Table 11: Two-population outcomes (exploratory). Win rate over the last 50,000 games; “best member” is the best individual in each pool at the end, against the 2015 policy.

condition	seeds	larger side wins cross-play	larger side’s pool, best member	smaller side’s pool, best member	runs where only one side’s pool learned
symmetric control (273 v 273)	6	0.62 (range 0.01–1.00); larger side ahead in 4/6	−2.35	−2.30	3/6
2:1 capacity, common σ	6	0.02 (range 0.00–0.75); larger side ahead in 2/6	−4.71	+0.25	4/6
2:1 capacity, matched step norm	6	0.95 (range 0.00–0.99); larger side ahead in 5/6	−1.21	−4.76	3/6

A.10 Alternative export rules, tested prospectively

Preregistered ([results/export/PREREGISTRATION.md](#)) and run on fresh seeds: 12 Slime Volleyball control runs and 48 discmix control runs. Because the exported member never feeds back into training, every rule is applied to the same stored populations, so each run is a paired comparison. Table 12 gives the Slime Volleyball result, Table 13 the discmix result judged against outsiders.

Table 12: Export rules on the same population snapshots, 12 fresh Slime Volleyball control runs. Level and declines: held-out score of the exported member against the 2015 baseline. Only the 16-peer tournament was tested; the other rows are described.

rule	games per export	level	declines	rank in population	final vs zoo GA
streak (Ha’s rule)	0	−2.33	1.98	56	−1.26
tournament, 4 peers	256	−2.31	1.97	55	—
tournament, 16 peers	1,024	−2.16	1.37	46	−1.25
tournament, 64 peers	4,096	−2.03	0.72	37	—
random member	0	−2.50	2.12	61	—
best member (oracle)	—	−1.93	0.61	2	−1.05

Table 13: Export rules in the discmix game, judged by outsider strength (exact expected score against the other runs’ populations at the same λ and snapshot).

λ	streak	tournament, 16 peers	best member (oracle)	tournament higher	rank: streak / tournament
0.00	+0.082	+0.189	+0.364	9/12	50 / 27
0.25	+0.027	+0.135	+0.292	12/12	58 / 31
0.50	+0.009	+0.065	+0.244	11/12	62 / 44
0.75	+0.002	+0.029	+0.159	10/12	66 / 49

A.11 The lab: cyclic skill, another archive, another topology

The three experiments of §3.9, each preregistered in the repository before its first run ([results/lab](#), [results/qd](#), [results/neat](#)) and analysed once. The lab’s GA reproduces the study’s three kernels bit for bit, and a NEAT genome that encodes the study’s network plays bit-identical games, so the numbers are comparable with the main text’s.

Table 14: The discmix game: cycling, the export rule and the time-ordered archive used as a test, by λ (12 runs per cell, exact expected scores).

λ	cyclic triads within runs (control / test)	exported rank in pool of 128	ρ (streak, strength)	decline: exported / best	archive as test vs control, δ (p)
0.00	0.0% / 0.0%	52	+0.13	0.00 / 0.00	-0.44 (0.068)
0.25	1.4% / 1.7%	56	+0.13	0.00 / 0.00	-0.29 (0.242)
0.50	5.7% / 6.2%	61	+0.08	0.00 / 0.00	-0.47 (0.052)
0.75	13.5% / 13.3%	63	+0.07	1.78 / 0.08	+0.06 (0.843)

Table 15: An archive organised by behaviour, used as a test, in Slime Volleyball, against the control and the time-ordered archive on the same seeds.

outcome	niche archive	control	δ (p)	time-ordered archive	δ (p)
final champion, held out	-1.21	-0.79	-0.11 (0.671)	-0.90	-0.15 (0.551)
best champion, held out	-0.60	-0.51	-0.24 (0.347)	-0.49	-0.06 (0.843)
checkpoints above parity	0.14	0.18	-0.23 (0.352)	0.14	+0.04 (0.875)
mean score, last 100,000 games	-1.41	-1.01	-0.26 (0.291)	-1.03	-0.17 (0.514)
learned to rally	10/12	10/12	—	10/12	—

Table 16: The behaviour-organised archive in the discmix game, by λ .

λ	niche vs control, δ (p)	niche vs time-ordered archive, δ (p)	occupied cells (of 64)	cyclic triads: control / time-ordered / niche
0.00	-0.32 (0.198)	+0.07 (0.799)	40	0.0% / 0.0% / 0.0%
0.25	+0.15 (0.551)	+0.39 (0.114)	40	1.4% / 1.7% / 1.7%
0.50	-0.06 (0.843)	+0.38 (0.128)	34	5.7% / 6.2% / 5.0%
0.75	+0.28 (0.266)	+0.11 (0.671)	34	13.5% / 13.3% / 13.0%

Table 17: NEAT as a fifth family next to the four of §3.7.

family	runs	learned to rally	reached parity	final (held out)	spread vs 2020 GA	best final	checkpoints above parity	median cross-run Elo
control (Ha 2020 GA)	6	6/6	6/6	-0.15 $\pm$ 0.66	1.00 $\times$	+0.41	0.26	+371
generational GA (Ha 2015)	6	5/6	5/6	-2.00 $\pm$ 2.00	3.05 $\times$	-0.09	0.03	+143
self-play ES	6	4/6	2/6	-2.08 $\pm$ 2.29	3.49 $\times$	+0.50	0.07	+206
archive as test, full span	6	5/6	4/6	-1.01 $\pm$ 2.03	3.08 $\times$	+0.35	0.18	+449
NEAT	12	0/12	0/12	-4.84 $\pm$ 0.01	0.02 $\times$	-4.82	0.00	-406

A.12 Every condition

Tables 18–21 give every metric for every condition, each condition’s comparison with the control, the within-run tournaments and the timing of the phase change.

Table 18: Every condition: endpoint and best checkpoint (held out), level, volatility and drawdown of the last 100,000 games, share of checkpoints above parity, median first-parity time.

condition	runs	final (held out)	peak (held out)	mean, last 100k	volatility	drawdown	above parity	median first parity
control (Ha 2020 GA)	6	-0.15 $\pm$ 0.27	+0.32 $\pm$ 0.06	-0.32 $\pm$ 0.23	0.67	0.60	26%	192k (6/6)
archive as test, full span	6	-1.01 $\pm$ 0.83	-0.55 $\pm$ 0.86	-1.06 $\pm$ 0.78	0.51	0.52	18%	195k (4/6)
archive as parent, p=0.25	6	-4.10 $\pm$ 0.74	-3.94 $\pm$ 0.90	-4.14 $\pm$ 0.70	0.12	0.24	2%	365k (1/6)
archive as parent, p=0.50	1	-4.84 $\pm$ —	-4.84 $\pm$ —	-4.85 $\pm$ —	0.01	0.06	0%	— (0/1)
archive as parent, full span	3	-4.84 $\pm$ 0.00	-4.83 $\pm$ 0.01	-4.84 $\pm$ 0.00	0.01	0.05	0%	— (0/3)
generational GA (Ha 2015)	6	-2.00 $\pm$ 0.82	-0.68 $\pm$ 0.83	-1.61 $\pm$ 0.70	0.63	0.70	3%	345k (5/6)
self-play ES	6	-2.08 $\pm$ 0.94	-1.56 $\pm$ 0.99	-2.46 $\pm$ 0.93	0.21	0.18	7%	398k (2/6)
sigma = 0.05	3	-0.17 $\pm$ 0.23	+0.34 $\pm$ 0.07	-0.29 $\pm$ 0.15	0.69	0.64	26%	150k (3/3)
sigma = 0.20	3	-1.66 $\pm$ 1.60	-1.44 $\pm$ 1.70	-1.76 $\pm$ 1.54	0.38	0.36	11%	290k (2/3)
population 32	1	-4.86 $\pm$ —	-4.84 $\pm$ —	-4.85 $\pm$ —	0.01	0.05	0%	— (0/1)

Table 19: Each condition against the control, exact two-sided Mann–Whitney.

condition	metric	difference	Cliff’s δ	exact p
archive as test, full span	final_holdout	−0.859	−0.28	0.485
archive as test, full span	late_mean	−0.745	−0.22	0.589
archive as test, full span	volatility	−0.157	−0.22	0.589
archive as test, full span	drawdown	−0.083	−0.11	0.818
archive as test, full span	above_parity	−0.087	−0.33	0.370
archive as parent, p=0.25	final_holdout	−3.951	−0.94	0.004
archive as parent, p=0.25	late_mean	−3.828	−0.94	0.004
archive as parent, p=0.25	volatility	−0.557	−0.83	0.015
archive as parent, p=0.25	drawdown	−0.366	−0.67	0.065
archive as parent, p=0.25	above_parity	−0.245	−0.94	0.004
archive as parent, full span	final_holdout	−4.688	−1.00	0.024
archive as parent, full span	late_mean	−4.528	−1.00	0.024
archive as parent, full span	volatility	−0.659	−1.00	0.024
archive as parent, full span	drawdown	−0.550	−1.00	0.024
archive as parent, full span	above_parity	−0.263	−1.00	0.024
generational GA (Ha 2015)	final_holdout	−1.852	−0.67	0.065
generational GA (Ha 2015)	late_mean	−1.299	−0.78	0.026
generational GA (Ha 2015)	volatility	−0.046	+0.00	1.000
generational GA (Ha 2015)	drawdown	+0.103	+0.00	1.000
generational GA (Ha 2015)	above_parity	−0.230	−0.83	0.015
self-play ES	final_holdout	−1.933	−0.33	0.394
self-play ES	late_mean	−2.145	−0.33	0.394
self-play ES	volatility	−0.464	−0.72	0.041
self-play ES	drawdown	−0.420	−0.83	0.015
self-play ES	above_parity	−0.197	−0.83	0.013
sigma = 0.05	final_holdout	−0.018	−0.22	0.714
sigma = 0.05	late_mean	+0.026	−0.44	0.381
sigma = 0.05	volatility	+0.017	+0.00	1.000
sigma = 0.05	drawdown	+0.034	+0.11	0.905
sigma = 0.05	above_parity	−0.003	−0.11	0.905
sigma = 0.20	final_holdout	−1.509	−0.33	0.548
sigma = 0.20	late_mean	−1.445	−0.44	0.381
sigma = 0.20	volatility	−0.294	−0.44	0.381
sigma = 0.20	drawdown	−0.238	−0.33	0.548
sigma = 0.20	above_parity	−0.150	−0.78	0.095

Table 20: Within-run checkpoint tournaments.

condition	runs	ρ (Elo, training time)	cyclic triads	undecided pairs
control (Ha 2020 GA)	6	+0.74	1/499 (0.2%)	10.2
archive as test, full span	6	+0.79	0/451 (0.0%)	12.8
archive as parent, p=0.25	6	+0.35	17/224 (7.6%)	20.8
archive as parent, p=0.50	1	+0.02	0/15 (0.0%)	26.0
archive as parent, full span	3	+0.06	9/29 (31.0%)	31.0
generational GA (Ha 2015)	6	+0.60	2/485 (0.4%)	12.3
self-play ES	6	+0.69	3/424 (0.7%)	14.3
sigma = 0.05	3	+0.91	0/239 (0.0%)	11.3
sigma = 0.20	3	+0.62	1/175 (0.6%)	15.7
population 32	1	+0.01	2/6 (33.3%)	29.0

Table 21: Phase-change timing per condition.

condition	runs	reached long rallies	internal transition (median, range)	first parity (median, range)	lag (median)
control (Ha 2020 GA)	6	6/6	120k (55k–415k)	192k (85k–440k) (6/6)	58k
archive as test, full span	6	5/6	150k (115k–190k)	195k (145k–260k) (4/6)	62k
archive as parent, p=0.25	6	1/6	80k (80k–80k)	365k (365k–365k) (1/6)	285k
archive as parent, p=0.50	1	0/1	never	never (0/1)	—
archive as parent, full span	3	0/3	never	never (0/3)	—
generational GA (Ha 2015)	6	5/6	205k (160k–420k)	345k (290k–480k) (5/6)	105k
self-play ES	6	4/6	372k (320k–490k)	398k (385k–410k) (2/6)	70k
sigma = 0.05	3	3/3	120k (80k–250k)	150k (145k–405k) (3/3)	65k
sigma = 0.20	3	2/3	228k (215k–240k)	290k (275k–305k) (2/3)	62k
population 32	1	0/1	never	never (0/1)	—
<i>reference run (1 run, real environment)</i>	1	1/1	<i>104k</i>	<i>172k</i>	<i>68k</i>

B The compiled environment

The environment, the evolved policy and the 2015 baseline were transcribed statement for statement into a numba-compiled module, including behaviour that looks like a bug but changes trajectories (for example, observations are not refreshed on a frame in which a point is scored). Three deviations are deliberate: the random-number generator family, the way two distinct population members are drawn, and the floating-point details of the network’s forward pass. The last can change a trajectory only by flipping the sign of a network output, because the game reads only whether each output is positive.

Per game. Driven from one stream of serve velocities, both implementations agree on every ball and agent position, every rally end, every score and every episode length in all 200 paired games (304,812 steps), across 4 populations chosen to exercise different parts of the state space (Table 22). The compiled version is 26 times faster on one core.

Per trajectory. Continuing the reference run’s committed population snapshot at 324,000 games for another 176,000 games, 2 compiled continuations put 54–60% of their checkpoints above parity, against 44% for the reference environment over the same games (Table 23). Independent continuations diverge because the algorithm is stochastic; they land in the same band.

Per distribution. The reference run’s transition times and best score lie inside the compiled control’s seed distribution (§3.2, Table 2).

Table 22: Bit-level validation of the compiled environment.

scenario	paired games	identical score	identical length	identical trajectory	max abs deviation	env steps compared
random vs random	50	50/50	50/50	50/50	0	31,971
champion vs random	50	50/50	50/50	50/50	0	60,465
champion vs champion	50	50/50	50/50	50/50	0	62,376
champion vs 2015 baseline	50	50/50	50/50	50/50	0	150,000

Table 23: Continuations of the reference run’s population snapshot.

continuation	games added	final score	checkpoints above parity
compiled, resume_s900	176,000	−2.82	19/35
compiled, resume_s901	176,000	+0.41	21/35
reference environment	176,000	+0.01	78/176

C Reproducing the results

Every number in this paper is regenerated from the run files by the repository’s scripts; with the pinned requirements installed, the sequence is

```
python test_repo.py                # self-test, incl. bit-level checks
python validate_fastvolley.py      # per-game validation
python run_experiments.py          # the single-population matrix
python run_asymmetric.py           # the two-population block
python train_ga_selfplay.py --resume # the reference run, slimevolleygym
python eval_reference.py
python analyze_matrix.py --holdout  # metrics and held-out re-scoring
python coevolution_analysis.py --within --across --proxy
python reexport.py                 # promotion rules
python resume_fast.py              # compiled continuations
python make_tables.py && python make_figures.py
paper/build.sh
```

The analyses added after the matrix, on data already on disk, and the preregistered follow-ups (each runner writes its run files, each analysis is the one fixed in the experiment’s preregistration):

```
python yardsticks.py               # final champions vs the zoo policies
python transitivity_fine.py         # transitivity at 5,000-game spacing
python proxy_heldout.py            # export check, re-scored held out
scripts/replicate.sh               # the replication’s runs
python replication.py               # results/replication
python run_lab.py && python lab_analysis.py      # results/lab
python run_qd.py && python qd_analysis.py        # results/qd
python run_neat.py && python neat_analysis.py    # results/neat
python neat_explore.py && python neat_explore.py --summary # exploratory
python run_export.py && python export_analysis.py # results/export
```

Runs whose files exist are skipped, so every runner can be resumed after an interruption. Each analysis script records the hash of every raw file it read and of its own output; `make_tables.py -check` fails if any table or number in the paper or the README no longer follows from the files on disk, and a continuous-integration job runs it, with the self-test, on every change. Rerunning the analysis reproduced the committed files: the reference curve and the compiled continuations byte for byte, the held-out scores, the within-run tournaments, the export-rule check and the promotion-rule analysis in every reported value. Every change to the design after the first run started is recorded, with what was known at the time, in the decision log shipped with the data; it includes the archive-as-test bug, its rerun and the comparison of the two.